BHARATHWAJ RAMANATHAN

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Education

University of Illinois Urbana-Champaign — Aug 2024 – Dec 2025
MEng, Mechanical Engineering | GPA: 3.91/4.00 — Champaign, IL

PSG College of Technology — Jul 2020 – May 2024
BE, Mechanical Engineering | GPA: 8.59/10.00 — Coimbatore, India

Experience

Vinjamuri Innovations — Mar 2026 – Present
Mechanical Design Engineer | Composite Structures — New Rochelle, NY

- Built a nonlinear **hybrid solid-shell FEA** model of a **Type III COPV** in **ANSYS 2025 R2**, predicting burst pressure within **14%** of a confirmed **1080 bar** test by simulating **3-step autofrettage** at **593 bar** for high-pressure hydrogen storage.
- Validated **7-ply CFRP/GFRP** stress distribution in **ANSYS ACP** using CNC winding data and confirmed structural safety against **ISO 11119-2** requirements at up to **350 bar** service pressure through residual stress and post-autofrettage load sharing analysis.
- Support the commercialization of **two patents for high-pressure industrial gas storage** through engineering, manufacturing, validation, and supplier planning.

Yantra Systems — Jan 2026 – Present
Product Development Engineer | Electromechanical Systems — Champaign, IL

- Led development of **EasyBlot**, an automated western blot platform that reduced hands-on experimental time from **18 hours to 1 hour**, improving throughput and reproducibility for research workflows.
- Designed and validated an automated **6-pump reagent transfer and fluid handling system**, integrating mechanical hardware, custom **2-layer PCB**, **ESP32**-based controls, and fluid routing into a functional prototype.
- Collaborated with researchers to define requirements, conduct technical demonstrations, manage vendor sourcing, and guide iterative prototype refinement.

University of Illinois Urbana-Champaign — Jan 2025 – Dec 2025
Graduate Researcher | Thermal Systems & CFD Simulation — Champaign, IL

- Designed and evaluated **BCC lattice cold plate architectures** for electronics thermal management, performing **30 CFD simulations** across geometry variants to quantify heat transfer and pressure drop tradeoffs; reliability-based design optimization identified an optimal configuration with a **maximum temperature of 311.52 K**, validated within **0.67%** of surrogate model predictions.
- Developed MATLAB-based **Gaussian Process surrogate models** to predict maximum temperature and pressure drop across lattice design variables, reducing design iteration time by **100+ hours** and enabling rapid design-space exploration.

Vinjamuri Innovations — Jun 2025 – Aug 2025
Mechanical Design Engineer Intern | Composite Pressure Vessels — New Rochelle, NY

- Performed nonlinear FEA on a patented **Type III composite hydrogen vessel** across **160–350 bar** operating pressures, evaluating stress and deformation behavior to establish safe operating margins and support structural design decisions.
- Developed a simulation-ready CAD model and modeled filament-wound composite layups in **ANSYS ACP**, defining laminate architecture and evaluating load transfer between the metallic liner and composite overwrap under high-pressure loading.

Prenac Tools — Dec 2023 – May 2024
Mechanical Engineering Intern | Structural Design & Manufacturing — Coimbatore, India

- Reduced material replacement costs by **30%** by redesigning a collapsible industrial pallet in **SolidWorks**, validating a **1-ton** load capacity through FEA and geometry improvements targeting structural durability and reuse cycles.
- Fabricated and load-tested the prototype through sheet metal cutting, welding, and **GD&T**-controlled assembly per **ASME Y14.5**, using sim-vs-test discrepancies to refine modeling assumptions and drive design improvements.

Projects

Immersion-Cooled EV Battery Module — Aug 2024 – Dec 2024
SolidWorks · ANSYS Icepak

- Achieved **46%** reduction in peak cell temperature and reduced thermal variation from **23.7 °C** to **7.0 °C** by designing and validating an **immersion cooling system** for a 9-cell cylindrical battery module using **ANSYS Icepak**.
- Selected **EG 50-50** coolant after evaluating six fluid candidates based on thermal performance and pressure drop, supporting cooling system component selection and design optimization while improving thermal uniformity.
- Compared cold plate and immersion cooling architectures using temperature distribution, pressure drop, and coolant performance metrics to guide system-level thermal design decisions.

Technical Skills

CAD & Design: SolidWorks, Fusion 360, Creo Parametric, nTopology, GD&T (ASME Y14.5), Tolerance Analysis, DFMA
Thermal & Fluids Simulation: ANSYS Fluent, ANSYS Icepak, SolidWorks Flow Simulation, SimScale, thermal management systems
Structural & Composite Simulation: ANSYS Mechanical, ANSYS ACP, Abaqus, nonlinear FEA, composite layup modeling, pressure vessel analysis
Simulation & Modeling: MATLAB, Python, Gaussian Process surrogate modeling, reliability-based design optimization
Manufacturing & Prototyping: Sheet Metal Design, Rapid Prototyping, Prototype Build and Validation
Electronics & Controls: KiCad, custom PCB design, ESP32